

Toward Simulation-Based Automatic Evaluation of Human-Perceived Traits in Robotic Agents

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Abstract—@TODO

Index Terms—Human-Robot Interaction, Human Perception, Automatic Evaluation, Physical AI Safety, Robotic Agents

I. INTRODUCTION

Robotic agents are moving from controlled laboratory settings toward human-centered environments, where they are expected not only to complete tasks, but to do so in ways that are safe, legible, and socially acceptable. As embodied or physical AI systems become more capable, the need for reliable benchmarks becomes increasingly urgent: researchers, developers, and users need principled ways to compare which robotic agent performs better in a given context.

A similar transition has already occurred in the evaluation of large language models. Early benchmarks mainly focused on task accuracy, while more recent evaluation frameworks attempt to quantify subjective and human-facing qualities of interaction. For example, Chatbot Arena ranks models through crowdsourced pairwise human preferences, MT-Bench evaluates multi-turn conversational quality using human-aligned judges, and HELM argues for multi-metric evaluation beyond accuracy, including robustness, fairness, bias, toxicity, calibration, and efficiency. These benchmarks show that, when AI systems are deployed in open-ended human contexts, performance cannot be reduced to objective correctness alone.

Robotics faces an even harder version of this problem. In Human-Robot Interaction (HRI), the success of an agent depends not only on whether it completes a task, but also on how its behavior is perceived by humans. A robot may be mechanically efficient while appearing unsafe, unaware, intrusive, hesitant, or socially inappropriate. These subjective impressions are difficult to measure, but they are central to real-world adoption: a robot that is not perceived as trustworthy, competent, or predictable may fail even when its low-level control policy is technically effective.

This work addresses the need for reproducible evaluation methods that connect objective robot behavior to human-perceived interaction traits. We propose a framework for estimating subjective traits of a robotic agent from objective measurements collected in a purpose-built simulation environment. By sampling robot behavior across controlled scenarios and aggregating measurable signals into human-centered dimensions, our goal is to make HRI evaluation more

systematic, comparable, and actionable during the design and benchmarking of robotic agents.

II. BACKGROUND

Surveying takes a lot of time and requires a lot of money. Consequently, it is not feasible to obtain reliable data when developing a robotic project

Currently, the state of the art methodology to evaluate how robots are perceived by humans heavily leans on surveys, like GODSPEED and NARS.

- Research on safety evaluation and decision algorithm of automatic driving based on artificial intelligence [1]
- Safety Assessment of Collaborative Robotics Through Automated Formal Verification [2]
- Safety Evaluation of Robot Systems via Uncertainty Quantification [3]

III. METHODOLOGY

We measure objective metrics in a simulation environment based on MuJoCo. We simulate different scenarios, and gather a specific set of objective metrics. We then aggregate these metrics into 4 different core metrics:

- perceived dexterity
- perceived safety
- perceived social awareness
- impression.

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IV. EXPERIMENTAL SETUP

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V. RESULTS

VI. CONCLUSION

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TABLE I
MACRO-METRIC SCORES BY EVALUATOR GROUP (MEAN $\pm$ STD, SCALE 0–100).

Metric	Auto Evaluator	Human (Random Pop.)	Human (Robotics St.)
Perceived Dexterity	67.7 $\pm$ 9.3	70.5 $\pm$ 12.2	54.6 $\pm$ 9.6
Perceived Safety	68.2 $\pm$ 9.6	66.7 $\pm$ 14.0	51.4 $\pm$ 9.6
Perceived Social Awareness	64.6 $\pm$ 9.1	63.4 $\pm$ 8.5	45.9 $\pm$ 7.2
Impression	72.8 $\pm$ 8.9	73.2 $\pm$ 11.8	57.2 $\pm$ 7.4
Composite	68.4 $\pm$ 4.3	68.4 $\pm$ 6.4	52.3 $\pm$ 3.8

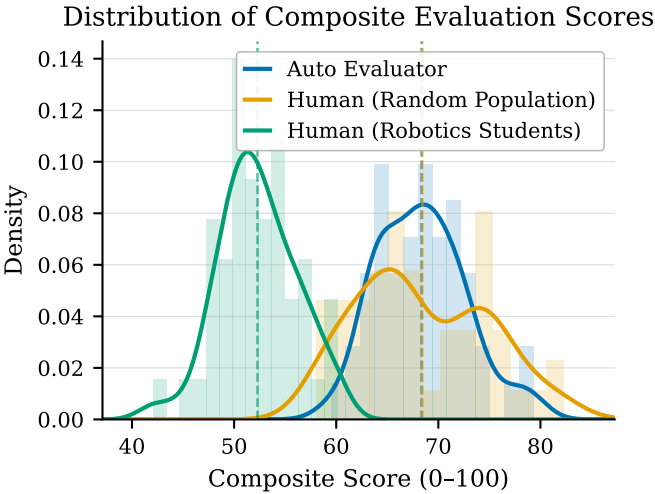


Fig. 1. Composite evaluation scores for the three evaluator groups. Dashed vertical lines mark group means.

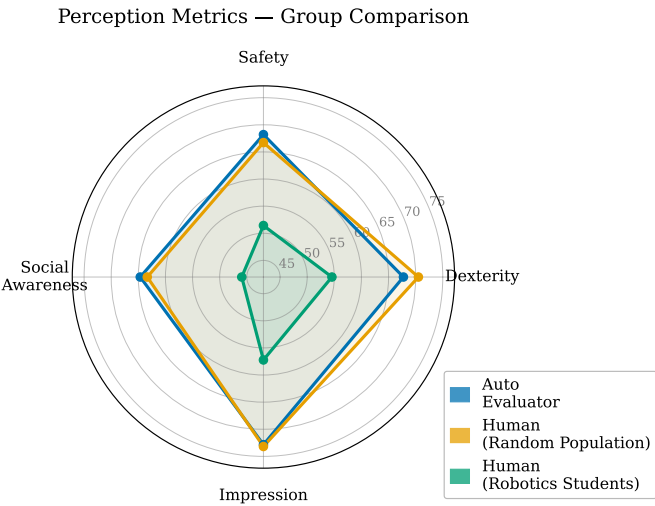


Fig. 2. Spider chart of mean macro-metric scores per evaluator group.

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